

Bhaskar Garg

📞 +91 82228 42347
✉ bgarg@ee.iitr.ac.in
🌐 www.linkedin.com/bhaskar/

EDUCATION

Indian Institute of Technology Roorkee(IIT Roorkee) India
B.Tech. Electrical Engineering (Agg percent:86%) 2018 – 2022

EXPERIENCE

Intel Technologies Bangalore, India
PD Methodology Engineer June 2022-Present

- TFM(Tool Flows and Methodology) owner for a critical IP consisting of more than 100 partitions ensuring quality execution adhering to schedules.
- Implementing **LLMs**(Large Language Models) in the BackEnd Methodologies, Experimenting with various possible use cases like **auto analysis of design reports** on various stages of Flow.
- Engineered an **automated VLSI design flow**, leveraging RTL to GDSII processes, and significantly enhanced chip design efficiency using Synopsys Fusion Compiler and TCL scripting, increasing **design process efficiency to 5x**.
- Developed a flow to port memory views across tech nodes to enable PnR work in early stage of projects, target node was **intel foundry** and parent project was in **TSMC node**.
- Implemented advanced automation for **repeater delay analysis**, optimizing signal integrity and timing across multi-node technologies, utilizing Python for **cross-technology analysis**.
- Directed Performance, Power, and Area (**PPA**) **optimization** experiments to push benchmark results in 18A tech node.

INTERNSHIPS

BHEL-IITR Consultation Project Haridwar, India
Industrial IoT Expert July 2021 – June 2022

- ATMEGA2560 based PCB design to handle 24 industrial sensors.
- Python GUI to read and control the machine at BHEL. ([PCB](#))

University of Windsor Windsor, Canada
MITACS Research Scholar May 2021 – July 2021

- Capacitive Design Sensors simulation on COMSOL.
- Result comparison for different Gas sensor Topologies. ([Report](#))

HMV Drives Roorkee, India
Research Intern Jan 2021 – May 2021

- Developed an IoT based motor controlling system for Farmers.
- Whatsapp Operated system to give a feel of actual assistant. ([Video PoC](#))

SKILLS

- 🔧 Python · TCL · Arduino · Xilinx Altium · · ESP · Eagle
- 📦 Cadence · MATLAB · Synopsys Fusion Compiler · Genus · Innovus
- 🌐 Proficiency in Basic PnR flows like Placement, CTS, Routing debug and analysis.

COLLEGE PROJECTS

- [Remote Covid Care Solutions for Doctors of Institute Hospital.](#)
- [AIR Quality Monitoring System using FPGA SPARTAN-3E.](#)
- BackPain Management Solutions using electronics wearable.

AWARDS

- **Distinguished Invention Award** for implementing Multi Agentic(LLM) workflow in Back End PD Flow.(Award is given if an invention is kept as a Trade secret, instead of making it publicly accessible by Patent)
- **Patent Granted** for "System and Method for monitoring and Controlling a Plurality of Field Devices"
- Gold Medal, Inter IIT Tech Meet'19.
- Prototype to Product Award by Design Innovation Centre, IITR.
- Winner at Hult Prize, IITR organised by Bill Clinton Foundation.

EXTRA-CURRICULAR

- U-19 National Level **Chess** Player
- Vice President of Electrical Engineering Student Society, 2021
- Volunteer at **Sanskrit** Club, IITR
- Member, IEEE Student Society, IITR